Section, at locations satisfactory to the Ministry Representative,

(c) appoint and maintain a Qualified Coordinator, meeting the qualifications of a "qualified coordinator" as described in the Workers Compensation Act and the Regulation, and notify the Ministry Representative and all Employers working within the Area of Responsibility of any changes to the appointment. The duties of the Qualified Coordinator include:

(i) to perform the functions of the "qualified coordinator" as described in the Regulation,

(ii) to ensure the coordination of health and safety activities for the Area of Responsibility, including establishing Joint Committees and holding meetings and maintaining records thereof,

(iii) ensure all Employers and Workers are informed of hazards created, and

(iv) ensure that the hazards are addressed throughout the duration of the work activities,

(d) ensure all other Employers designate and at all times maintain not less than one Qualified Person to be responsible for the Employer's health and safety activities connection with the Work, including administering the Safety Requirements,

(e) ensure periodic crew safety meetings are held, and minutes kept, to inform the Workers of any known or reasonably foreseeable risks on the Area of Responsibility and the actions to be taken to eliminate or minimize those risks,

(f) ensure appropriate first aid resources meeting Regulatory requirements are always present at the Area of Responsibility,

(g) ensure the availability of first aid attendants and that they are only assigned duties that allow them to respond immediately without any resulting risk to the workers or the public.

(h) ensure safe access for delivery of equipment and material,

(i) create, implement and maintain an AoR-Specific Safety Plan, and submit that Plan to the Ministry Representative for review and acceptance, a minimum of seven (7) days prior to Work commencing. The submission shall include a Quality Control check sheet, identifying that all items in this SS 135.05.02(i) have been addressed and are compliant with all Safety Requirements.

The AoR-Specific Safety Plan must be provided to all contractors and Employers within the Area of Responsibility, and must explain and demonstrate how safety will be managed within that Area, without limiting the generality of the foregoing, the Plan shall provide:

(i) the name of the person(s) designated to supervise the Works,

(ii) the name of the Qualified Coordinator,

(iii) the name(s) and certification level(s) of first aid attendant(s) and a means of summoning an attendant,

(iv) a system for identifying foreseeable and known hazards that may affect Workers and Employers,

(v) a system for communicating hazards to Employers and Workers and ensuring Employers notify the Prime Contractor of undertakings likely to create a hazard,

(vi) that the quality management systems of all Employers operating within the Area of Responsibility include provisions for the review of the Safety Requirements and the Contractor's compliance with those requirements,

(vii) a set of construction procedures designed to address hazards and protect the health and safety of Workers,

(viii) emergency response procedures which address potential hazards, or impacts to the AoR, during regular and after hours work activities, including but not limited to:

(A) first aid response and coordination,

(B) the provision of first aid supplies, equipment and facilities

(C) identification of, and directions to, medical aid, including the means of transportation of injured workers, and

(D) the method of notifying workers, Employers and others the notification methods for evacuating the AoR and identification of appropriate muster stations,

(ix) a drawing showing the project layout, first aid locations, emergency transportation provisions and the evacuation marshalling station,

(x) measures to ensure:

(A) regular inspections are carried out,

(B) all Workers and visitors receive an occupational health and safety orientation,

(C) regular site safety and toolbox meetings are held, including by other Employers,

(D) any Incidents are reported, investigated, and reviewed in accordance with SS 135.05.02(j) and SS 135.05.02(k),

(j) immediately notify the Ministry Representative of any Incident that the Prime Contractor becomes aware of that:

(i) resulted in serious injury to or the death of a worker,

(ii) involved a major structural failure or collapse of a building, bridge, tower, crane, hoist, temporary construction support system or excavation,

(iii) involved the major release of a hazardous substance, or

(iv) was an incident required by the Regulation to be reported. or

(v) materially impacts the movement of traffic or presents a safety concern for the travelling public.

(k) ensure any of the above Incidents, and any others resulting in medical aid, property damage in excess of five thousand dollars ($5,000.00), or involving members of the public are reviewed and investigated, and that incident reports are provided to the Ministry Representative within 72 hours from the time of occurrence, and

(l) when requested by the Ministry Representative, shall prepare and provide a statistical summary, on Ministry form H-1024 (link below), of all health and safety records applicable to the Area of Responsibility and covering the period requested by the Ministry Representative.

https://www.th.gov.bc.ca/forms/getForm.aspx?formId=783

(m) follow the Notice Process specified in SS 135.08 whenever an unsafe or harmful condition or act occurs in the Area of Responsibility or on an Adjacent or Nearby Worksite,

135.05.03 Base Safety Program – Notwithstanding the limitation relating to the requirement for initiating and maintaining an occupational health and safety program set out in the Regulation, the Prime Contractor will submit, to the Ministry Representative, a Base Safety Program a minimum of seven (7) days prior to the commencement of work, or as requested by the Ministry Representative.

The Prime Contractor will initiate and at all times maintain, at the Area of Responsibility, the Base Safety Program which will be designed to prevent injuries and occupational diseases within the contemplation of the Safety Requirements, describe how the Prime Contractor will meet the requirements of SS 135.05.02 and, without limiting the generality of the foregoing, will describe how the Prime Contractor will ensure that:

(a) the requirements of an occupational health and safety program as stipulated in the Safety Requirements will be satisfied, including the establishment and maintenance of a Joint Committee,

(b) the activities of all Employers, Workers and Other Persons at the Area of Responsibility and relating to occupational health and safety will be coordinated,

(c) reasonably practicable measures are taken to establish and maintain a system or process to ensure compliance with the Safety Requirements in respect of the Area of Responsibility, including ensuring regular health and safety inspections are carried out by each Employer,

(d) the Prime Contractor, all Other Authorized Parties, all Workers, Employers, and Other Persons

(i) are made aware of all known or reasonably foreseeable health and safety hazards to which they may be exposed on the Area of Responsibility,

(ii) notify the Prime Contractor of any operations they plan to undertake that may create new hazards, and

(iii) are instructed in the terms of the Safety Requirements,

(e) complete and accurate reports, records and documents relating to occupational health and safety activities at the Area of Responsibility, or related to the Work, will be established, maintained, reviewed and assessed on an ongoing basis,

(f) safety equipment and apparel are properly used and worn by Workers and Other Persons, as required pursuant to the Safety Requirements, and

(g) a first aid program in compliance with the Safety Requirements is established, implemented and maintained at all times on the Area of Responsibility.

135.05.04 Base Safety Program Implementation – The Prime Contractor will determine and control the manner in which the Base Safety Program is established, initiated, and maintained and the manner in which the instructions, directions, rules, policies and procedures contained therein are carried out.

135.05.05 Prime Contractor Obligations to Cooperate with Adjacent or Nearby Sites – If another party is the prime contractor in connection with another Adjacent or Nearby Worksite or Area of Responsibility, the Prime Contractor:

(a) will continue to undertake and perform the obligations of a Prime Contractor in the Area of Responsibility at all times in accordance with the terms of the Safety Requirements,

(b) will advise the other party in writing of the name of the Qualified Coordinator, and of any change in the Qualified Coordinator, designated to be responsible for the health and safety activities on or in connection with the Area of Responsibility, and any activities in the Adjacent or Nearby Worksite,

(c) will co-operate with the other party so as to accommodate any requests that may be issued by the other party to cooperate the health and safety activities on the Adjacent or Nearby Worksite, and

(d) may request the cooperation and accommodation of prime contractors on Adjacent or Nearby Worksites to ensure the health and safety activities in the Area of Responsibility are compliant with the Safety Requirements.

135.05.06 Copies of Records – When requested, the Prime Contractor shall provide to the Ministry Representative copies of any records mentioned in SS 135.05.

135.06 Contractor

135.06.01 Contractor's Deemed Agreement – The Contractor, by entering into the Contract, agrees to, for the purposes of the Workers Compensation Act and the Regulation, the designation of the Prime Contractor and the Area(s) of Responsibility made by the Ministry pursuant to SS 135.04.01 to SS 135.04.03.

135.06.02 Commencement of Work – The Contractor will, prior to commencement of the Work, deliver to the Prime Contractor, in writing:

(a) the name of each of the Qualified Persons in connection with the Site, and

(b) a description of the works the Contractor will be undertaking in the Prime Contractor's Area of Responsibility,

(c) a copy of all Notices of Project.

135.06.03 Contractor's Responsibilities – the Contractor will, in connection with the Work and the Site:

(a) observe, abide by and comply with the Safety Requirements, including providing any first aid resources, necessary for the Contractor's operations, which are beyond those provided through the Prime Contractor,

(b) ensure all required Notices of Project have been submitted to the Workers' Compensation Board in accordance with the Workers Compensation Act and the Regulation, and that other agencies are notified as specified in the Safety Requirements,

(c) prominently post and at all times maintain on the AoR all Notices of Project in accordance with the Workers Compensation Act and the Regulation,

(d) ensure that the person or persons to be appointed Qualified Person meet the qualifications of a "qualified person" as described in the Workers Compensation Act and the Regulation,

(e) designate and at all times maintain not less than one Qualified Person to be responsible for health and safety activities of the Contractor in connection with the Work, including, without limiting the generality of the foregoing, administering the Safety Requirements, as they applies to the Contractor,

(f) deliver written notice to the Prime Contractor of any change in the appointed Qualified Person referenced in SS 135.06.02(a),

(g) participate in the Joint Committee and ensure that the Contractor's Workers attend periodic safety meetings established by the Prime Contractor including the Prime Contractor's site safety orientation,

(h) establish and participate in Other Joint Committees and periodic crew safety meetings to inform the workers of any known or reasonably foreseeable risks on the Site and the actions to be taken to eliminate or minimize those risks, and maintain minutes of those meetings,

(i) establish and maintain at a location on the Site, or at a location off the Site that is satisfactory to the Prime Contractor, accurate records, including supporting documents, relating to the Contractor's implementation, operation, and maintenance of the Safety Requirements,

(j) make copies of the Safety Requirements readily available at the Site,

(k) provide and maintain the Site in a manner that ensures the health and safety of persons at or near the Site,

(l) ensure that the Prime Contractor is given the information known to the Contractor that is necessary to identify and eliminate or control hazards to the health or safety of Workers and Other Persons at the Site,

(m) ensure that the Prime Contractor, Workers, Employers and Other Persons are made aware of all known or reasonably foreseeable health and safety hazards to which they are likely to be exposed by the Work on the Site,

(n) do everything that is reasonably practicable to establish and maintain a system or process that will ensure compliance with the Safety Requirements with respect to the Site,

(o) immediately notify the Prime Contractor, Employers, Workers, Suppliers and Subcontractors of the Contractor, Other Authorized Parties, and Other Persons of any hazard created by virtue of overlapping or adjoining work activities of two (2) or more

Employers and ensure that the hazards are addressed throughout the duration of such activities,

(p) immediately notify the Prime Contractor and Ministry Representative of the occurrence of any Incident that:

(i) resulted in serious injury to or the death of a worker,

(ii) involved a major structural failure or collapse of a building, bridge, tower, crane, hoist, temporary construction support system or excavation,

(iii) involved the major release of a hazardous substance, or

(iv) was an incident required by the Regulation to be reported. or

(v) materially impacts the movement of traffic or presents a safety concern for the travelling public.

(q) ensure any of the above Incidents, and any others resulting in medical aid, property damage in excess of five thousand dollars ($5,000.00), or involving members of the public are, reviewed, investigated and that incident reports are provided to the Prime Contractor within seventy-two (72) hours from the time of occurrence.

(r) comply with any directive related to occupational health and safety matters issued by the Prime Contractor, and

(s) provide to the Prime Contractor:

(i) the name of the person designated to supervise the Contractor's Workers,

(ii) the name of the Qualified Person(s),

(iii) copies of safety meeting records and toolbox talks,

(iv) copies of worksite inspections,

(v) any and all notices, directives and inspection reports which have been delivered to the Workers' Compensation Board by the Contractor or which the Contractor has received from the Workers' Compensation Board, Ministry of Mines, or other regulatory body,

(vi) a list of known site hazards and of any operations the Contractor is planning on undertaking that are likely to create a hazard, and updates as hazards change,

(vii) immediate notification of any incidents, including near misses,

(viii) incident investigation reports,

(ix) a written summary of the actions taken by the Contractor to coordinate, in conjunction with the Prime Contractor, the occupational health and safety activities of the Employers, the Workers, and the Other Persons within the Site,

(x) statistical summaries, on Ministry form H-1024 (link below), as necessary to respond to a request initiated by the Ministry Representative,

https://www.th.gov.bc.ca/forms/getForm.aspx?formId=783

and

(xi) such other information as the Prime Contractor may request.

135.06.04 Payment Conditional Upon Documentation – The delivery of any payments by the Ministry to the Contractor under the Contract is conditional on the prior submission to the Prime Contractor by the Contractor, of the documentation required under this Section.

135.07 Other Authorized Parties

135.07.01 Health & Safety Terms and Conditions – All Other Authorized Parties shall comply with the health and safety related terms and conditions of their respective agreements with the Ministry, and upon request shall provide a copy of those terms and conditions to the Ministry Representative and the Prime Contractor.

135.07.02 Inconsistencies – Where the terms of their specific agreement appear to be inconsistent with the requirements of the Prime Contractor for any Area of Responsibility, the Other Authorized Party shall cooperate with the Prime Contractor and the Ministry in resolving any such inconsistency.

135.07.03 Default Terms and Conditions – Should the Other Authorized Party's agreement with the Ministry not contain any health and safety coordination responsibilities, the Other Authorized Party shall be subject to any health and safety requirements established by the Prime Contractor.

135.08 Notice Process

135.08.01 Employers to Provide Notice – Any Employer who becomes aware of an unsafe or harmful condition or act shall provide notice to the Prime Contractor in that Area of Responsibility.

135.08.02 Notice of Unsafe or Harmful Condition or Act – If the Prime Contractor, in connection with the Area of Responsibility, determines, in its reasonable discretion, that:

(a) any Employer or its Workers has failed to comply with the Safety Requirements, and

(b) the failure creates or may create an unsafe or harmful condition in the Area of Responsibility that poses a danger of injury or death to Workers,

the Prime Contractor, may issue a Notice of Unsafe or Harmful Condition or Act to the applicable Employer, stipulating in reasonable detail the basis for the issuance of the Notice of Unsafe or Harmful Condition or Act, and the Prime Contractor, will deliver a copy of the Notice of Unsafe or Harmful Condition or Act to the Ministry Representative.

135.08.03 Exclusions – For the purposes of this SS 135.08, the terms "Employer" and "Workers" will not include the Prime Contractor's Subcontractors or Suppliers, or their respective employees, workers, servants, agents or invitees, as the Prime Contractor is directly responsible for the actions of those parties.

135.08.04 Ministry to Issue Stop Work Order – Upon receipt of the Notice of Unsafe or Harmful Condition or Act, the Ministry Representative will issue to the applicable Employer a Stop Work Order at the location in the Area of Responsibility that is specified in the Notice of Unsafe or Harmful Condition or Act.

135.08.05 Deemed Reliance – The Ministry will be deemed to have relied on the terms of and reasons for the issuance of the Notice of Unsafe or Harmful Condition or Act, notwithstanding any subsequent investigation or inquiry of the matter by the Ministry which the Ministry may but will not be obligated to undertake.

135.08.06 Rectification – If the Prime Contractor and the Employer referenced in SS 135.08.02 fail to agree on the reasons for the issuance of the Notice of Unsafe or Harmful Condition or Act, then within one day of the issuance of the Notice of Unsafe or Harmful Condition or Act, the directive of the Prime Contractor to rectify the failure to comply with Safety Requirements, will be immediately implemented.

135.08.07 Conclusion of Unsafe or Harmful Condition or Act – When the failure to comply with the Safety Requirements stipulated in the Notice of Unsafe or Harmful Condition or Act has been rectified, to the reasonable satisfaction of the Prime Contractor, by the Employer to whom it was given to, the Prime Contractor, will immediately:

(a) withdraw the Notice of Unsafe or Harmful Condition or Act by endorsing on a copy thereof confirmation that the failure to comply with the Safety Requirements has been rectified.

(b) deliver the endorsed copy of the Notice of Unsafe or Harmful Condition or Act confirming the cancellation thereof to the applicable Employer and to the Ministry Representative.

135.08.08 Ministry to Issue Resume Work Order – The Ministry Representative will issue a Resume Work Order on receipt of the endorsed copy cancelling the Notice of Unsafe or Harmful Condition or Act.

135.08.09 Delay and Impacts – If the Prime Contractor is delayed in performing the Work as a result of being required to issue the Notice of Unsafe or Harmful Condition or Act and the Work is stopped pursuant to an Stop Work Order under SS 135.08.04, then the Completion Date will be extended by the number of days equal to the delay, as determined in the sole discretion of the Ministry Representative, and the Prime Contractor will not be entitled to any compensation, in damages or otherwise, including without limiting the generality of the foregoing, any claim for lost profits or other consequential loss, including business loss, as a result of or in connection with the issuance and operation of any Notice of Unsafe or Harmful Condition or Act or the issuance and operation of any Stop Work Order pursuant to this Section.

135.08.10 Withhold Payments – Notwithstanding any other term of this Section or of the Contract, the Ministry may withhold any payment due to the Contractor under the Contract if the Prime Contractor has issued, in connection with the Site, a Notice of Unsafe or Harmful Condition or Act in accordance with this Section, and the Ministry may continue to withhold such payment until such time as the Notice of Unsafe or Harmful Condition or Act has been cancelled and the Resume Work Order has been issued.

APPENDIX 135-A

PRIME CONTRACTOR'S NOTICE OF UNSAFE OR HARMFUL CONDITION OR ACT (the "Notice")

This Notice has been issued by the Prime Contractor to:

Name of Contractor: _____

Address: _____

The following unsafe or harmful condition or act was observed at the following location and time:

Description of unsafe or harmful condition or act: _____

_____

Location: _____ Date: _____ Time: _____ AM/PM

The Contractor named in this Notice is required to correct the above referenced unsafe or harmful condition or act prior to performing any more work activity at the location identified above.

Prime Contractor: _____

Signature: _____ Date: _____ Time: _____ AM/PM

Receipt acknowledged by the Contractor:

Signature: _____ Date: _____ Time:_____ AM/PM

The Ministry Representative received a copy of this Notice on: Date: _____ Time: _____ AM/PM

On the basis of this Notice, the Ministry Representative will issue a "Stop Work Order" in connection with the unsafe or harmful condition or act as identified on this Notice.

The Prime Contractor will, if satisfied that the unsafe or harmful condition or act has been rectified cancel and withdraw this Notice.

Date of cancellation/withdrawal: _____ Time: _____ AM/PM

Prime Contractor - Signature: _____

The Ministry Representative will issue an "Resume Work Order" when in receipt from the Prime Contractor of this Notice endorsed to indicate that the unsafe or harmful condition or act has be rectified.

DISTRIBUTION: Original: Contractor Copy: Ministry Representative File Copy: Prime Contractor

CONSTRUCTION SITE SAFETY SIGN	SP135–01



NOTE:
CONTRACTOR PROVIDES COMPANY NAME AND PHONE NUMBER OF THE APPROPRIATE CONTACT PERSON.

SPECIFICATIONS

SIGN SIZE	(MINIMUM) 2440 mm x 610 mm (8'–0" x 2'–0") (MINIMUM) 2440 mm x 305 mm (8'–0" x 1'–0")
MATERIAL	10 mm PLYWOOD OR .081" SHEET ALUMINUM
COLOUR	BLACK MESSAGE ON WHITE REFLECTORIZED BACKGROUND
TEXT SIZE	100 mm AND CENTERED HORIZONTALLY

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

SECTION 145

GENERAL REQUIREMENTS FOR HIGHWAY, BRIDGE, AND MARINE CONSTRUCTION

145.01 General Requirements – This Section covers the general Work requirements for highway, bridge, and marine construction.

The standards and materials specifications contained in the Standard Specifications for Highway Construction govern the specific requirements for the Work. Special Provisions provide additional specific requirements for the Work and supersede the Standard Specifications where there is conflict.

Specifications, standards, regulations and codes of other authorities named in full or by acknowledged abbreviations in the Glossary of Terms of the Standard Specifications or the Contract Document Package shall be adopted by reference.

Such reference in whole or in part by the necessary alphanumerical designation from the latest edition or revision is intended to fully detail the requirements of the Work.

145.02 Reference Information – Reference information for the Work, in the form of General Reference Documents and Specific Reference Documents, is identified in the Tender and the Contract, and is made available under the conditions specified therein.

145.03 Most Recent Metric Edition – Unless otherwise specified, any standard, specification, publication or similar document shall be the most recent edition, using metric units of measure, published on or before the date the Work is tendered or otherwise first offered.

Where an earlier edition is specified in these Standard Specifications, the reference will be in a form that includes the year of the specific edition that applies.

As an example, ASTM C136 would mean the current metric edition (i.e. as of 2020-11-01: ASTM C136M-19); ASTM C136-14 would specifically mean the 2014 edition.

Unless otherwise specified or implied in the context, where a standard or specification has both metric and imperial editions, the metric edition shall apply to the Work.

145.04 Scope of Work – The scope of Work is described in the Contract. No warranty is given or implied that any or all of this Work will be required or supplied.

145.05 Quantities and Dimensions – Quantities, dimensions, lines and elevations as shown on the plans and Drawings are designed neat lines. Approximate quantities entered in Schedule 7 – Approximate Quantities and Unit Prices are quantities approximated using the lines and dimensions shown in the Contract Document Package and other relevant factors, and may vary from quantities derived from the Drawings. Actual payment quantities will be measured, calculated, or otherwise determined in accordance with the Contract.

145.06 Unit Prices – Unit Prices in the Schedule of Approximate Quantities and Unit Prices shall be complete and contain all the elements of cost involved, provision of all things, superintendence and profits. The price shall stand independent of any other Unit Price Item on the Schedule of Approximate Quantities and Unit Prices.

145.07 Pre-construction (Post–Award) Meeting – The Contractor shall prepare the required documentation in accordance with the Contract, then advise the Ministry Representative, who will determine the pre-construction meeting's location and time and notify the Contractor. The meeting shall be held reasonably soon after Award and is intended to afford both parties an opportunity to establish relationships and to gain an enhanced understanding of the project and each party’s plans and expectations.

145.08 Mobilization – Mobilization consists of the necessary Work and operations including, but not limited to, the movement of personnel, equipment, supplies, and incidentals to the Site, the establishment of offices, camps, and other facilities necessary to undertake the Work and all other Work Items and operations which must be initiated and finished as part of completion of the Work.

145.08.01 Payment for Mobilization – Where the Contract includes a payment Item for mobilization, the Lump Sum Price for mobilization includes any or all of the related expenses incurred for mobilization, demobilization and any re-mobilization not covered under the Work Items which must be initiated and finished as part of the Work. The Lump Sum Price for mobilization will be paid in increments as the Work progresses.

If the Contract does not contain a payment Item for “Mobilization”, all costs associated with mobilization, demobilization and re-mobilization will be included in payment for Items having such costs.

Payment for mobilization, if applicable, will be made at the tendered Lump Sum Price as follows:

(a) The Contractor will be paid any portion of the mobilization Lump Sum Price exceeding five percent (5%) of the Tender Price within sixty (60) days after the Actual Completion Date or on the final progress estimate (whichever occurs first).

(b) That portion of the mobilization Lump Sum Price not exceeding five percent (5%) of the total Tender Price, will be paid to the Contractor according to the following breakdown:

(i) Seventy-five percent (75%) of this portion when the Contractor has submitted, in an acceptable form, to the Ministry Representative each of the following that is required under the Contract

(A) Construction Schedule,

(B) Cash Flow Projection Schedule,

(C) Traffic Management Plan,

(D) Quality Control Plan,

(E) Construction Environmental Management Plans,

(F) Sediment and Drainage Management Plan,

(G) Base Safety Plan and Area of Responsibility Safety Plans, as applicable,

and the value of the Work completed on bid Items other than mobilization exceeds two percent (2%) of the Tender Price.

(ii) fifteen percent (15%) of this portion of the mobilization Lump Sum Price will be paid prorated on a monthly basis for the percentage of Work completed as determined by the Ministry Representative, but contingent upon the Ministry Representative receiving acceptable construction schedule and cash flow updates and all required or requested quality control records or reports, and

(iii) the final ten percent (10%) of this portion of the mobilization Lump Sum Price when all completion documentation ha**s** been submitted by the Contractor as required by the Ministry, all equipment has been removed from the Site, and the Site has been cleaned up to the satisfaction of the Ministry Representative.

The payments from the Lump Sum Price, as set out above, will be full compensation for mobilization, demobilization and re-mobilization, regardless of the number of times the Contractor mobilizes.

145.09 Authority of the Ministry Representative – The Ministry Representative has the authority to accept or reject the Contractor's Work in accordance with the conditions set out in the Contract. Until the Work meets the requirements of the Contract and has been accepted by the Ministry Representative, the Ministry is not obligated to make payment for Work done by the Contractor. The Ministry Representative shall take all necessary measurements and calculations to determine the final quantities for final payment.

145.10 Power, Telephone Lines and Other Utilities – Moving of existing power and telephone lines, poles and other utilities from their current to their final design locations, unless specified as being part of the Work, shall be performed by third parties at the Ministry's expense.

If the Contractor's operations require the temporary removal or relocation of lines or poles, such removal or relocation shall be entirely the responsibility of the Contractor and costs of any interim utility relocations shall be to the Contractor's account. Utility owner(s) must be contacted prior to doing any utility work.

The Contractor shall be responsible for any damage to poles, lines or utilities caused by the Contractor's construction operations, and shall repair and make good the same or bear the expense of making good and repairing the same.

The Contractor shall take the necessary precautions to protect existing utility services during the performance of the Work and shall be responsible for any losses or damages caused by the Contractor.

Locations of the utilities shown on the Drawings are approximate and may not be complete. The Contractor shall be responsible to ensure the locations of the utilities are complete and shall notify the Ministry Representative of any discrepancies between the Drawings and the Contractor's findings.

145.11 Permits – The Contractor shall give all requisite notices in connection with the Work to the proper authorities and shall procure at the Contractor's expense all permits, licences, etc., of every description necessary for the construction and completion of the Work.

The Contractor shall deliver or make available all original certificates to the Ministry Representative for all or any part of the Work for which such certificates may be required in connection with performing the Work.

145.12 Ministry Inspection of Work

145.12.01 General – All materials and each part or detail of the Work shall be subject to inspection by the Ministry Representative or designated inspector.

The Contractor shall be held strictly to the true intent of the Contract in regard to quality of materials, work practice and the diligent execution of the Contract.

The Ministry Representative shall be given access to all or any part of the Work, and to the preparation, fabrication or manufacture of the materials to be used. The Contractor shall furnish all information and assistance as is required to make a complete, true and detailed inspection.

The Ministry Representative may reject any part or parts of the Work which do not comply with the requirements of the Contract. The Ministry's decision will be accepted as final.

Inspection may extend to all or any part of the Work and to the preparation, fabrication or manufacture of the materials to be used. The Ministry may station others on the Site to

report on the progress of the Work and the manner in which it is being performed.

The Ministry Representative will report to the Contractor on materials furnished or Work performed that fails to fulfil the requirements of the Contract. Ministry costs for re-inspection and re-testing for repaired and replaced Work items that originally did not meet Contract requirements shall be at the Contractor's expense.

Inspection or its lack shall not relieve the Contractor from any obligations to furnish acceptable materials or to provide completed construction Work that complies with the Contract.

The Contractor shall furnish for the approval of the Ministry Representative, at proper times, all shop and installation drawings as requested by the Contract or which the Ministry Representative may deem necessary.

In accepting drawings submitted by the Contractor, the Ministry Representative will assume no responsibility as to the correctness of such drawings and the Ministry Representative's examination is only to ensure general compliance with the Contract.

145.12.02 Hold Points and Witness Points – Hold Points and Witness Points (as defined below) are required at critical times for the Ministry Representative and the Engineer(s) of Record (EOR)/Professional(s) of Record (POR) to review crucial work processes.

(a) Definitions

(i) **Hold Points** are verification points identified by the Ministry Representative beyond which Work may not proceed until:

(A) the Contractor notifies the Ministry Representative,

(B) the Ministry Representative, or the Professional of Record, performs mandatory verification; and

(C) the Ministry Representative gives notice to the Contractor to proceed.

(ii) **Witness Points** are identified points in the process where the Ministry Representative may review, witness, inspect methods or processes of Work. The activities however may proceed.

(b) **Hold Points** – Hold Points may be specified throughout these Special Provisions, the SP Appendices, the Drawings, and in the Standard Specifications.

The Contractor shall provide the Ministry Representative seven (7) days' notice in advance of a Hold Point to allow the Ministry Representative to provide adequate notice to the EOR.

Additional Hold Points may be added by the Ministry Representative at any time during the Contract, by written notice to the Contractor.

(c) **Witness Points** – Witness Points may be specified throughout these Special Provisions, the SP Appendices, the Drawings, and in the Standard Specifications.

The Ministry Representative will provide the Contractor with notice of any additional Witness Points as the Contract progresses. The Contractor will provide the Ministry Representative with one (1) clear days' notice of the commencement of the work requiring witnessing.

145.13 "Ministry's Instruction" and "Contractor's Declaration"

145.13.01 "Ministry's Instruction" – "Ministry's Instruction" form (H0202) will be issued by the Ministry Representative to the Contractor in order to:

(a) describe a portion of the Work which may not be covered specifically in the Contract.

(b) confirm instructions which may affect the method of carrying out the Work or payment.

(c) confirm verbal agreements made with the Contractor's representative.

(d) respond to the Contractor's representation on "Contractor's Declaration" form (H0203).

145.13.02 "Contractor's Declaration" – The Contractor is to issue the "Contractor's Declaration" form (H0203) to the Ministry Representative in order to:

(a) make a declaration of acceptance or objection to contents of the "Ministry Representative's Instruction".

(b) formally present the Contractor's opinions, problems or requirements for clarification.

145.14 Protection and Restoration of the Work – The Contractor shall secure and protect the Site, all Work, and the surrounding infrastructure from loss, injury or damage until the issuance of the Completion Certificate.

If the Contractor fails to secure and prevent any loss, injury, or damage to the Work, the Contractor shall rebuild, repair, replace, and restore at its expense any portion of the Work so lost, injured or damaged.

The Ministry may reimburse the Contractor for any injury or damage caused by an event not foreseen and not reasonably foreseeable by the Contractor and beyond the control and not resulting from any fault or negligence of the Contractor including but not restricted to acts of God, acts of a Public Enemy, acts of the Ministry, extraordinary action of the elements and unavoidable slides.

145.15 Materials Supplied by the Contractor – The Contractor shall provide all materials necessary for the orderly completion of the Work with the exception of the materials supplied by the Ministry as listed in the Special Provisions.

Further requirements for asphalt materials are contained in SS 502**.**

All materials supplied under the Contract shall:

- be new,
- conform to the requirements of the Contract, and
- be subject to inspection and approval by the Ministry prior to incorporation in the Work.

145.15.01 Ministry Inspection of Materials – Includes material certificates review, fabrication inspection, sampling and testing of materials.

The Contractor shall provide to the Ministry, material or mill certificates two weeks prior to shipping of the materials supplied by the Contractor. Such material or mill certificates shall include the material properties that demonstrate the material's compliance with the requirements of the Contract.

At the request of the Ministry Representative, the Contractor shall submit representative samples of materials at no charge to the Ministry.

The Contractor shall notify the Ministry Representative of the sources of materials supplied by the Contractor, and obtain the right for the Ministry to enter upon the premises of the material supplier or fabricator to carry out such inspection. Such notification shall be given sufficiently far in advance of materials delivery dates to enable the Ministry Representative to make inspection of the material at the source. Insufficient notice will be cause for the Ministry to reject the materials without compensation.

The Ministry will not be responsible for any delays to the Contractor's operations where the Contractor fails to give sufficient advance notification to the Ministry Representative to enable the Ministry to carry out the inspection before the scheduled shipping dates.

145.15.02 Materials or Components Sourced from Outside British Columbia – If materials or components to be incorporated into the Work are sourced from outside the Province then:

(a) the Contractor will be charged for all travel time, transportation, and board and lodging costs incurred by the Ministry or its designate for out-of-province quality assurance and inspection of such materials and components,

(b) the Contractor will be charged for the costs incurred by the Ministry in inspecting replacement materials and components or re-inspecting reworked materials and components which did not meet the requirements of the Contract, and

(c) the Ministry will offset the costs incurred by the Ministry from any payments due to the Contractor

145.15.03 Not Used

145.15.04 Alternative Materials – Brands, species and manufacturer's names specified in the Contract are approved for use in the performance of the Work. No alternative materials will be considered unless approved in advance by the Ministry Representative in writing.

Savings as a result of using alternate materials will be evaluated using SS 125 Value Engineering Proposals.

145.15.05 Recognized Products List – The Ministry has reviewed a number of products for use in highway applications and publishes a "Recognized Products List", indicating which products are considered generally acceptable for particular purposes. Contractors are encouraged to review the list to aid in determining which products they will supply, but are cautioned that the products listed, although generally acceptable, may not meet Contract specifications.

The Categories "Proven, "Tentative" or "Approved" in the Recognized Products List are an indication of the product's record of use and purpose. Products in any category are "recognized" and may be used as long as the product meets Contract specific specifications.

Unless so specified in the Contract, Materials provided are not required to be amongst those on the Recognized Products List.

The Recognized Products list is available on-line at:

https://www2.gov.bc.ca/gov/content/transportation/transportation-infrastructure/engineering-standards-guidelines/recognized-products-list

145.16 Materials Supplied by the Ministry – Ministry supplied materials will be procured and paid for by the Ministry by purchase order to the Purchasing Commission, who on instruction from the Ministry Representative notifying of the Contractor's requirement for the material, shall arrange for release of the material to the Contractor in accordance with the purchase order.

Separate purchase orders will be issued for each class of material.

Materials ordered for a specific Site shall not be transferred to any other Site without the written consent of the Ministry Representative.

Ministry supplied and owned materials will be used only for Ministry Work.

145.16.01 Delivery Points – Each purchase order will specify points of delivery. The Contractor shall be

responsible for all costs that may arise such as unloading from truck or rail car, hauling to the Site, stockpiling, dunnage, and reloading.

The costs to transport all materials to the Site will be incurred by the Contractor at its own cost and will be considered incidental and no separate payment will be made for this portion of the Works.

Should the Contractor request delivery be made to an alternative point other than the nearest specified point of delivery or in any manner of delivery other than specified by the Purchasing Commission, the additional cost for this delivery shall be charged to the Contractor's account.

145.16.02 Delivery and Storage of Materials – When the Ministry Representative advises the Contractor of the name of the supplier(s), the Contractor shall be responsible at no cost to the Ministry for:

(a) determining requirements and ordering far enough in advance that work is not delayed.

(b) acceptance of all materials; careful unloading and handling; hauling and delivery from delivery point to an interim storage facility and to the Site; storage and security; prompt turn-around of tank or shipping cars, trucks or delivery containers.

(c) prompt payment of all demurrage or rental charges on cars, freight, handling, delivery, storage or any other charges arising after the initial receipt of material at the delivery point.

(d) ensuring that proper accounting is kept of all materials received.

The Contractor shall notify the Ministry Representative of the materials delivery schedule and allow sufficient time for the Ministry Representative to check quantity and quality before the Contractor takes delivery. The Contractor shall not take delivery of any material without the presence of the Ministry Representative.

The Contractor shall report to the Ministry Representative any materials damage or discrepancies between the quantities received and the quantities shown on the bills of lading immediately upon materials arriving at the delivery point. If damage or deficiencies are not so reported, it will be assumed that the materials arrived in good order.

The Contractor shall provide a secure storage acceptable to the Ministry Representative for all materials supplied by the Ministry. The Contractor shall, in advance of receipt of the shipment of the Materials, notify the Ministry Representative of the storage facilities locations for their inspection and acceptance.

The materials shall be kept in the designated storage immediately upon receipt of the materials. The Contractor shall not remove any materials from the storage locations, except for incorporation in the Work, without authorization from the Ministry Representative.

The Contractor shall be solely responsible for replacement of any materials due to loss, damage, or improper use prior to acceptance into the final Works.

If asphalt materials are hauled directly from the delivery point for immediate application to the roadway, the material will be measured by mass or volumetrically by the Ministry Representative at the time of delivery. The Contractor shall ensure that an acceptable means of measurement is available.

145.16.03 Distribution of Materials – To control distribution of Ministry supplied materials and to facilitate accounting:

(a) the Contractor shall arrange for delivery of the materials only after applying for and receiving authority to do so on "Authority to Arrange for Delivery of Construction Materials" form (H0313) which will be provided by the Ministry Representative. The requirements of the Contractor shall be indicated on Part "A" of the form. Approval from the Ministry Representative shall be indicated on Part "B" of the form. The Contractor may then arrange for delivery from the supplier. As delivery will be made only on receipt of the purchase order number and the serial number of the "Authority" form, the Contractor shall arrange with the Ministry Representative to immediately distribute appropriate copies of the "Authority" form.

Material suppliers will be required to notify both the Ministry Representative and the Contractor when materials are shipped, giving sufficient information to identify each and every shipment against an appropriate "Authority" form serial number and purchase order number.

(b) The use of the "Authority" form shall not in any way relieve the Contractor of any responsibility imposed under SS 145.16.02.

(c) If the Ministry Representative refuses to authorize the delivery of materials to the job, the Ministry Representative will send the refusal, with reasons in writing, to the Contractor.

(d) The Ministry may, at the completion of the Work, if materials are delivered in excess of job requirements through conditions beyond the control of the Contractor, take over the surplus material or grant the Contractor the option of purchasing the material on Site. The transaction will be completed only after the Contractor obtains for the Ministry Representative, a supplier's credit note for the total cost to the Ministry of the excess material purchased DDP (Incoterms© 2020: Delivery Duty Paid) at the delivery point.

Upon completion of the Work, the Contractor will deliver all unused materials, which have been purchased by the Ministry, to a location as specified by the Ministry Representative and shall dispose of all empty materials packaging that are not required for their original purpose at the disposal site. No payment will be made for this portion of the Work.

145.17 Operation of Pavement Brooms on Highway – No mechanical pavement broom shall be operated on any highway without a rotating amber flashing light in operation. The light shall be mounted on self-propelled brooms and upon the towing vehicle of trailer type brooms.

Traffic Control is to be in accordance with the *Traffic Control manual for Work on Roadways*. In addition, on extremely winding narrow roads an additional shadow vehicle shall be used ahead of the broom with a C-076 "Sweeper Working" sign on the front of the vehicle and a flashing arrow in bar mode.

145.18 Supply and Maintain Facilities

145.18.01 Working Office – When specifically required under the Special Provisions, the Contractor shall provide a suitable working office with lock-up for the sole use of the Ministry. The office shall meet the requirements of APPENDIX 145-A and be weatherproof, reasonably soundproof, and have ample window area and ventilation. Location of the office shall be determined by the Ministry Representative.

The Contractor shall regularly collect refuse and keep the office clean and properly maintained with heat and light.

The office and contents shall be for the use of the Ministry for the duration of the Work and may if necessary be used concurrently with the other inspection agencies.

Where service is available on-Site, through land-line or by satellite, the Contractor shall provide high-speed wireless internet for the Ministry office. If cellular coverage is not available or is poor at the office site, but land-line services are available nearby, the Contractor shall supply a private telephone line for the sole use of the Ministry Representative. Ministry's long distance calls will be paid for by the Ministry. Where cell coverage is poor, the Contractor may provide a cell signal booster, instead of a land-line, if it provides a reliable signal.

Payment for the Ministry's Site Office (when required) will be made at the Lump Sum price in the Contract.

Payment will include the provision of the office, furnishings, communications systems, maintenance throughout the construction period, and removal in its entirety.

Payment of the Lump Sum will be made as follows:

(a) 50% when the office is ready for occupancy;

(b) 40% paid monthly, prorated to the percentage complete of the Work, and

(c) the remaining balance after the office has been removed and the Site cleaned up to the satisfaction of the Ministry Representative.

145.18.02 Testing Facility – When specifically required by the Contract, the Contractor shall provide the Ministry with a material testing facility with a minimum of 12 m^2 floor space either in the form of a small trailer or the portion of a large trailer. The testing facility shall include power, light, heat, ventilation, plus basic furniture to permit testing and preparation of test records. The facility shall be stable, reasonably quiet and lockable.

Payment for the Ministry's Material Testing facility (when required) will be made at the Lump Sum price in the Contract

Payment will include the provision of the facility, furnishings, maintenance throughout the construction period, and removal in its entirety.

Payment of 50% of the Lump Sum will be made when the facility is ready for occupancy and the remaining 50% after it has been removed and the Site cleaned up to the satisfaction of the Ministry Representative.

145.18.03 Service for Ministry Facilities – Where the Ministry provides its own facilities on Site, the Contractor shall, for the duration of the Contract, provide light and heat and any other services requested by the Ministry Representative for same, and the Contractor will be compensated on a Force Account Basis.

145.18.04 Weigh Scale

(a) Except as provided in SS 145.18.04(b), the Contractor shall provide a truck weigh scale wherever measurement of materials by mass is required.

The weigh scale shall be an approved electronic weigh scale or platform type with remote balancing beam. The scale shall be of sufficient size and capacity to weigh, at a single pass, the largest, rigid-framed haulage vehicle to be used on the job and in any case shall be of not less than 13 t (tonne) capacity. The scale shall be capable of weighing all loads to within 20 kg of the true mass.

At the start of the Work, and as frequently as the Ministry Representative may deem necessary in order to ensure the accuracy, the scale shall be calibrated and checked at the Contractor's expense by an Inspector of Weights and Measures.

The remote balancing beam shall be enclosed in a sound, watertight, dust-free scale house, properly ventilated and insulated. Adequate heating must be provided when the shade temperature is below 16°C.

The scale house shall be provided with full length windows facing the scale platform, and windows shall also be provided at either end, so that both the Contractor's scaleperson and the Ministry Representative's scale checker can at all times see vehicles to be weighed, not only on the scale but also approaching and leaving.

The front window shall be sufficiently large and low that the whole scale platform is readily visible from the scale house. Immediately behind the balancing beam, a bench or table of sufficient length shall be provided to permit both scaleperson and checker to sit abreast in a position in which they can both write on the table and view the electronic scale or weigh beam and scale platform.

SS Drawing SP145-01 shows these requirements.

The Contractor shall supply and maintain, at no additional expense to the Ministry, portable washroom facilities at the weigh scale site.

Payment for the weigh scale and ancillary facilities shall be incidental to the Unit Prices of the materials being weighed.

(b) Where any aggregate is being paid by weight (i.e. per tonne) and the required quantity for the Item is less than 5,000 t, the Contractor will be permitted to use any scale that is certified and able to provide a printed ticket (e.g. loader bucket scale) in lieu of a stationary weigh scale.

Such alternative scales shall not be used to weigh aggregates into either concrete or asphalt mixes.

145.19 Weighing Materials – The Contractor shall supply, at no additional expense to the Ministry, a qualified scaleperson to carry out the necessary weighing which may be monitored by the Ministry. A weighing certificate prepared in triplicate, certified by the scaleperson, will be issued for each load. The Contractor will be paid only for such material as has been received at the point of delivery designated by the Ministry Representative.

The delivery receipts will bear the signature, initials or other approved marking of the issuing scaleperson and receiving checker.

If the Contractor has an automated scale and the following conditions are being met, the Contractor shall not be required to supply a scaleperson:

(a) All weighed material is being delivered to a single job site. If the Contractor is delivering to two or more job sites at the same time, the Contractor shall supply a scaleperson as detailed in the first paragraph.

(b) Truck operators do not require instruction.

(c) There are no overloads, tarp covering, spillage difficulties, or any safety problems.

(d) The scale platform is maintained by the Contractor to be clean and free from loose material at all times.

(e) All vehicles being scaled must be tared at least once a day.

In the event that any of the above conditions are not being met, in the opinion of the Ministry Representative, then the Contractor shall supply a qualified scaleperson as detailed in the first paragraph to carry out the necessary weighting and issuing of a weighing certificate, prepared in triplicate, for each load.

The Contractor will be paid only for such material as having been received at the point of delivery designated by the Ministry Representative.

The delivery receipts will bear the signature, initials or other approved marking of the issuing scaleperson and receiving checker and shall be submitted to the Ministry Representative within forty-eight (48) working hours from each load delivery for approval.

In the event the Ministry Representative determines a scaleperson is not required, all weighed vehicles must be tared by both the Ministry Representative and the Contractor together once a day. The Contractor must provide certified automated scales that will generate weigh tickets. The Contractor's scaleperson will be required to scale and record the weights of the material required for the project according to the requirements of the SS 502.04.04. The Contractor shall supply the Ministry Representative with the calibration record of the balance. The calibration record shall be done once a year or when needed.

At any time the Ministry Representative may direct that any loaded truck return to the scale and be reweighed in the presence of the Ministry Representative.

145.20 Not Used

145.21 Survey/Layout and Calculation of Quantities – In general and as detailed below, the Ministry will establish control line points for the Site, survey the original and excavated ground lines, and calculate quantities for payment. The Contractor shall lay out the Work, and assume full responsibility for the alignment, dimensions and elevations of each and every part of the work and their mutual agreement.

145.21.01 Ministry Responsibilities – The Ministry will provide a listing of any existing survey control line (primary) reference points to the Contractor.

The Ministry will survey the original ground line, and any subsequent material horizons, for determination of appropriate pay item quantities.

Where the Ministry has a pre-existing survey of the Site, the Ministry may use that survey as the original ground. If the Ministry elects to use such a survey, the Ministry Representative will notify the Contractor of such intent and give the Contractor an opportunity to view the cross-sections, contours, or Digital Terrain Model. Any areas of the survey or model that do not adequately reflect the Site at the start of construction will be identified by the Contractor or the Ministry and will be resurveyed by the Ministry prior to the Contractor starting work in any affected area of the Site.

Unless otherwise specified in the Contract, calculation of quantities will be on a neat line basis, as calculated from the design lines shown on the Drawings, adjusted to incorporate design changes and field-fits authorized by the Ministry Representative. Any material placed or excavated beyond the design lines is considered Unauthorized Work.

For items such as Type D excavation where the upper boundary of a cut cannot be determined until the completion of stripping, the design line will be revised to reflect the actual bottom of stripping line as determined by the Ministry's survey.

Quantities calculated from digital terrain models will be done using average-end-areas based on cross-sections extracted from the model or, at the Ministry Representative's option, by using prismoidal volumes between surfaces in the model.

Where agreed in writing between the Ministry and the Contractor, alternate methods of determining payment quantities may be employed.

The Contractor may be required to adjust its construction activities from time to time to allow the Ministry's measurement work to proceed unencumbered. The Contractor shall not proceed with the next stage of any construction until the Ministry has completed measurements of the previous work.

Quality Assurance and checking of layout may be performed by the Ministry Representative.

145.21.02 Contractor's Responsibilities – The Contractor shall perform all layout of the Work and all other survey work required to execute the Contract and construct the works in accordance with the Contract.

The Contractor shall be satisfied, before commencing any Work, as to the meaning, intent and accuracy of any control points, control lines and benchmarks established by the Ministry.

Should the Contractor discover or suspect any errors in any control points, control lines, benchmarks, and data provided by the Ministry, the Contractor shall at once discontinue the affected work until such errors are investigated by the Ministry Representative and, if necessary, rectified.

(a) Conventional Survey Layout – The Contractor shall be responsible for establishing all secondary control points and/or lines, all slope stakes, the establishment of line and grades for subgrade and the various granular aggregate layers, layout by line and grade of all structures, culverts, and underground utilities, and shall perform all other layout and measurement necessary for the proper execution of the Contract. Such layout shall include making incidental field adjustments, such as staking of embankments and culverts to match post-stripping ground lines and actual field drainage patterns.

The Contractor's survey layout shall conform, as a minimum, to the requirements outlined in APPENDIX 145-B – Survey Layout Services and Products.

(b) Machine Controlled Grading – The Contractor may utilize machine- controlled grading as a substitute for conventional grade staking under the following conditions:

(i) The equipment utilized shall be capable of meeting the Design vertical and horizontal tolerances.

(ii) The Ministry Representative may require the Contractor to revert to conventional staking methods at any point during construction if the machine-controlled grading is producing Unacceptable Work.

(iii) The Ministry will provide the Contractor the available electronic files of Design information in accordance with the terms specified in the Special Provisions without warranty with respect to the suitability for the purposes intended by the Contractor.

(iv) As a minimum the Contractor shall provide an orientation stake every 100 metres showing station, offset and grade.

(c) Calculations – All calculations necessary shall be performed by the Contractor and provided to the Ministry Representative at any time upon request. Upon request of the Ministry Representative the Contractor shall provide the Ministry Representative a copy of calculations prior to the start of earthwork operations.

(d) Accuracy of Layout and Construction – The staking of all works shall be of a sufficient accuracy and frequency for the Ministry to carry out its quantity measurements and quality assurance program.

On request of the Ministry Representative, the Contractor shall submit documentation to verify the accuracy of the layout work and the construction of the Work.

Unless a tighter tolerance is required for the satisfactory performance of the Work, the Work shall be so laid out and constructed that the as-built locations of the Work shall not differ from the design positions by more than the following amounts:

(i) Elements and their tolerances as specified in APPENDIX 145-B

(ii) Substructure elements – 12 mm horizontally and vertically.

(iii) Anchor groups – 6 mm in plan.

(iv) The elevation of bearing seats shall not differ from true elevations by more than 3 mm.

Any deviation beyond the above-mentioned tolerances shall be satisfactorily corrected by the Contractor at the Contractor's expense.

145.21.03 Payment for Survey/Layout – Except as noted below, survey layout is incidental to the Work and no separate payment will be made.

The Contractor shall provide any additional or revised survey layout for any design change directed by the Ministry Representative and will be paid for such additional survey layout. Compensation will be determined in accordance with GC 38.00 and any approved payments will be made from the Provisional Sum for Site Modifications.

145.22 Access to Site – The Contractor shall provide access to the Site as may be necessary, unless otherwise specified in the Special Provisions. No separate payment will be made for the provision of any access roads or incidental works in connection therewith.

The Contractor shall make arrangements with the property owners adjacent to and around the terminal site for the use of access and for the use of any storage areas the Contractor may require. The Contractor shall acquire such access and storage rights at the Contractor's expense.

In the event of roads being temporarily closed to the travelling public, the Contractor shall, at the Contractor's expense, provide, erect and maintain all requisite barriers, fences or other proper protection and must provide, keep and maintain such patrollers, lights and danger signals as may be necessary, or as may be required by the Ministry Representative to ensure safety to the public and to the workers.

The Contractor, during the Contractor's operations, must also provide and maintain reasonable road access and egress to property fronting along or in the vicinity of the work under contract, unless other means of road access exists. The Ministry Representative will be the sole judge as to what may be deemed reasonable road access.

145.23 Accommodation of Traffic – The Contractor shall cause as little inconvenience as possible to the travelling public during the Contractor's operations and shall erect and maintain proper and adequate barricades, traffic signs, lights and other traffic control devices as may be considered necessary, in the opinion of the Ministry Representative, for the safety of both workers and public traffic.

All barricades, traffic signs, lights and other control devices shall be provided at the Contractor's expense, and shall be erected in accordance with, and otherwise conform to the standards as set out in the Ministry's *Traffic Management Manual for Work on Roadways*, Special Provisions, and Standard Specifications.

145.23.01 Maintenance of Traffic – The Contractor shall comply with the requirements of the Transportation Act and amendments thereto and such other Acts, By-Laws, or Regulations as are in force for the governing and regulating of traffic or use of any road upon or over which it is necessary to do work or transport materials. The Contractor must arrange to keep roads open to traffic at all times and must cause as little inconvenience as possible to the travelling public.

145.23.02 Road Closure Notification – Road Closures will be in accordance with SS 194 and the Special Provisions.

All costs related to the above shall be for the Contractor's account.

145.24 Alterations to Drawings and Plans – It shall be understood that the Drawings and Plans represent the nature and general layout of the work to be executed and not necessarily the works exactly as they will be carried out. The Ministry Representative will, without invalidating the Contract, be at liberty to make any reasonable alteration or to furnish any additional or amended Drawings which do not change the type of construction.

Payment for any such altered Work will be at the rates set forth in the Schedule 7 Approximate Quantities and Unit Prices.

145.24.01 Adherence to Drawings – The various parts of the Work shall be built in strict accordance with the furnished Drawings and with any supplementary Drawings which may be furnished during the progress of the Work as occasion requires.

Special Provisions, special specifications, Standard Specifications and the Drawings/Plans are to be considered as explanatory of each other, but should anything appear in one that is not described in the other, or should any discrepancy appear, no advantage shall be taken of such omission or discrepancy and the decision of the Ministry Representative as to the true intent and meaning of the Special Provisions, special specifications, Standard Specifications and the Drawings in such instances shall be final and binding, it being clearly understood that the Contractor is to execute all the works requisite for the perfect completion of each and all of the several parts of the

Work according to the true intent and meaning of the Contract.

On all Drawings, in case of any discrepancy between the figured dimensions and the scaled dimensions, the figured dimensions shall govern.

145.25 Use of Explosives – In using explosives the Contractor shall exercise the utmost care so as not to endanger life or property and, whenever directed, the number and size of the charges shall be reduced. Before doing any blasting the Contractor shall ascertain if there are in the neighbourhood any railway, power or other lines, water or other pipes, etc., which are likely to be in danger of being damaged. The Contractor shall give to the owner of such lines, pipes or other works timely notice of the intention to blast.

Whenever possible electric batteries shall be employed for simultaneous blasting.

The storage of explosives shall be in accordance with the Explosives Regulations of the Canada Explosives Act and with any requirements of the representative of the Explosives Regulatory Division within the Mineral Technology Branch of the Minerals and Metals Sector of the Federal Department of the Natural Resources Canada.

145.26 Work in Pits or Quarries – The Contractor shall be responsible for compliance with appropriate safety, environmental, forestry, agricultural and other applicable regulatory agencies for all work related to development, operation and reclamation of pit or quarry.

145.26.01 Ministry Pit or Quarry – When operating in a Ministry pit or quarry the Contractor shall comply with all provisions of the Mines Act, the Health, Safety and Reclamation Code for Mines in British Columbia, and/or the WCB Occupational Health & Safety Regulation which are applicable to the work undertaken.

Prior to entering any Ministry Pit or Quarry the Contractor shall provide a completed Sand and Gravel/Quarry Notice of Work (Form H1258) to the Ministry Representative for each Ministry Pit or Quarry the Contractor intends to enter in, at, or about to mine fourteen (14) days before, including any seasonal activation**.**

145.26.02 Contractor Pit Manager – The Ministry may, at its sole discretion, require that the Contractor, or a party other than the Contractor, take responsibility to ensure that the work in the pit or quarry is done in compliance with the requirements of the *Mines Act* (RSBC 1996) and the *Health, Safety and Reclamation Code for Mines in British Columbia* (B.C. Reg. 126/94). When the Contractor is so required, the Contractor must provide the name of a person, or persons, who will act as the Pit Manager to the Ministry Representative prior to the commencement of work. This person must possess the qualifications established by the regulations or code, and attend daily at the pit or quarry when it is in operation.

When the role of Pit Manager is not the responsibility of the Contractor, the Contractor will co-operate with, and follow the direction of the acting or appointed Pit Manager.

145.26.03 Pit Supervisor – The Contractor shall appoint a properly qualified supervisor or shift boss in any Ministry pit or quarry where work is undertaken, in compliance with Section 26 of the *Mines Act*, and ensure that the supervisor or shift boss is certified in accordance with Section 1.13 of the *Health, Safety and Reclamation Code for Mines in British Columbia*.

145.26.04 Notice of Unsafe Condition – If the Contractor, in connection with the pit or quarry, determines, in its reasonable discretion, that any person, employer or worker has failed to comply with any safety regulation, Act, or code, or any applicable site practice or procedure that creates or may create an unsafe or harmful condition that poses a danger of injury or death, the Contractor will deliver a written notification of this failure to the Ministry Representative.

145.26.05 Use of the Pit or Quarry – When the Contractor is operating in a Ministry pit or quarry, the Contractor shall utilize the pit or quarry in accordance with the Ministry Pit Development Plan or obtain written approval from the Ministry Representative prior to deviating from the plan. Upon completion of the work, the Contractor shall re-grade all faces in the pit in accordance with the Ministry Pit Development Plan and/or an approved Pit Reclamation Plan.

Ministry pits or quarries may be used without charge, provided they are utilized, worked and maintained in accordance with the Standard Specifications and the Ministry's development plans. No warranty is given or implied that the source of aggregate material from the Ministry pits or quarries will meet requirements or requisite quantities.

The Contractor shall contact the Road and Bridge Maintenance Contractor a minimum of two (2) weeks before the date the Contractor anticipates the use of any Ministry pits or quarries.

The Contractor and the Road and Bridge Maintenance Contractor shall co-ordinate their activities and schedule usage of the pit among themselves and any other affected parties.

The Contractor, by entering an available source, will be deemed to have accepted all obligations, risks and costs associated with its use and security, just as if the pit or quarry was private.

No warranty is given that the source will meet the quality and quantity requirements. The Contractor, without any additional compensation, shall provide whatever selection, screening, crushing, mixing, blending or other processing of any kind may be required to produce aggregate meeting all

requirements for its intended use, including but not limited to:

(a) specified aggregate gradation,

(b) fracture count,

(c) aggregate quality requirements such as but not limited to those set out in SS 202.04 and Table 501-D of SS 501 as applicable, and

(d) the specified properties of any products into which the aggregate is to be incorporated.

The Contractor shall comply with the Ministry pit development plan and the Ministry Representative's directions regarding the location of material extraction within the pit by the Contractor.

The Contractor's use of Ministry pits is limited to the extraction of previously unexcavated material from the bank, for the supply of materials for use on this Contract.

The Ministry's Road and Bridge Maintenance Contractor has certain rights and obligations with respect to some Ministry pits, and the Contractor shall coordinate its activities with those of that contractor, where applicable, and with any other users, to the satisfaction of the Ministry Representative and the Ministry's District Manager, Transportation.

No separate payment will be made for clearing, grubbing, disposal or relocation of stockpiles, debris or contaminated materials, or for any other costs of site preparation, pit development, or access, or for any delay or other cost arising from the use of Ministry pits by others, and all costs thereof shall be covered in the prices for the Items under which payment is provided for the applicable materials.

145.26.06 Materials Sourced from Ministry Lands but Processed Elsewhere – Where the Contractor elects to source raw materials from a Ministry pit or quarry, but process those materials on private lands:

(a) The Contractor shall extract only as much raw material from the Ministry source as is necessary to produce the quantity of product required to be incorporated in the final Project, inclusive of staging and maintenance requirements, with a small additional contingency amount as approved by the Ministry Representative.

(b) Ministry-sourced materials shall not be used in any way to improve the private lands, including without limitation for such purposes as access development, maintenance, or stockpile site preparation, unless approved in advance by the Ministry Representative under any conditions the Ministry Representative may require.

(c) Any surplus raw or processed material remaining on the private lands at the end of construction shall:

(i) be returned by the Contractor to the Ministry source and stockpiled at a location determined by the Ministry Representative, all at the Contractor's expense; or

(ii) upon request by the Contractor and at the Ministry Representative's sole discretion, the Contractor may be offered the options of:

(A) of leaving the materials on the private land and providing a written agreement with the owner of the property which shall state that the Ministry will have free access to and exclusive use of the remaining materials for a period of twenty-four (24) months after the completion of the Contract; or

(B) purchasing the remaining material at a negotiated amount to be back-charged to the Contractor.

(iii) Where the volume of material remaining is less than the additional contingency amount determined pursuant to SS 145.26.06(a), the Ministry Representative will make the offers in SS 145.26.06(c)(ii). If agreement is not reached, SS 145.26.06(c)(i) will apply.

145.26.07 Private Pit or Quarry – Prior to supply of any material to the Site, the Contractor shall obtain and provide to the Ministry Representative written approval from the titled land owner, for the right to occupy and operate private pit or quarry, including appropriate permits.

Availability and use of the listed reserves or any other private aggregate source must be arranged by the Contractor. Prior to the removal of any aggregate material from an aggregate source, the Contractor shall provide a written notice from the owner authorizing the Contractor to remove aggregate material for use in the work.

145.27 Highway Right-of-Way – The Contractor shall confine its operations to those portions of the highway right-of-way and Licences to Construct indicated in the Contract as being available to the Contractor. Any land, property or water lot outside these boundaries required for construction yards, access roads, disposal or storage areas, or other operations of the Contractor, shall be acquired by the Contractor at no additional expense to the Ministry. The Contractor shall make arrangements for the use and compensation of such land with the titled land owner.

Work within the Site shall be confined to that necessary to construct the Project to the lines and grades specified. Excavation outside the design lines and grades to extract aggregates or for other purposes is prohibited unless authorized in advance by the Ministry Representative.

145.27.01 Not Used

145.27.02 Disposal Site Outside the Right-of-Way – Waste, or other materials specified or designated by the Ministry's Representative to be disposed of, such as, but not limited to clearing and grubbing materials, fences, old culverts, old concrete and asphalt, etc., shall be removed from the Site to a disposal site provided by the Contractor. All disposal sites shall be the responsibility of the Contractor.

The Contractor shall ensure that the disposal site complies with all Federal, Provincial, First Nation, Municipal and Regional by-laws and regulations prior to disposing of the material. Any fees incurred shall be paid by the Contractor.

The Contractor must provide documented proof to the Ministry Representative that the selected disposal site complies with the above requirements.

145.27.03 Access Roads to Private and Public Lands – The Contractor shall not block or impede the use of any private access roads or driveways during the Work of this Contract without the consent of the Ministry Representative.

The Contractor shall schedule any anticipated closure of an access with the property owner to minimize impact on the owner.

145.28 Protection of Land Subdivision Monuments, Roads and Property

145.28.01 Protection of Subdivision Monuments – The Contractor shall protect, preserve, and keep uncovered all land subdivision monuments or property marks during its construction activities which may cause disturbance to them. At points where construction activities will cover or destroy any land subdivision monuments or property marks, the Contractor shall restore them afterwards to the satisfaction of the Ministry Representative.

145.28.02 Protection of Survey Monuments – The Contractor shall be responsible for the preservation during construction of all geodetic benchmarks, survey monuments and property markers on the right-of-way. The Contractor shall use, at no additional expense to the Ministry, a British Columbia Land Surveyor to replace any survey monuments destroyed or damaged as a result of the Contractor's negligence. At locations where construction work will cover or destroy such markers, the Contractor shall not move or remove them until written direction is received from the Ministry Representative.

145.28.03 Protection of Roads – The Contractor shall ensure all equipment working within, or hauling material to and from, the Site does not damage or deposit material onto any part of an existing roadway. Materials spilled onto the public roadways or driveways opened to public traffic shall be cleaned up immediately. The Contractor has the full responsibility to repair any damage to existing highways, local roads and driveways caused by its construction equipment and/or operations.

The Ministry manages a Province-wide seasonal strength loss program that, in conjunction with engineering assessments, determines the timing and degree of axle load restrictions which will apply to Ministry roads outside the Limits of Construction. Such restrictions do not apply within the Limits of Construction, but the Contractor is responsible for any damage caused by its operations within that area.

145.28.04 Protection of Railway Property – The Contractor shall be wholly responsible for any damage to railway trackage or property on the railway company's right-of-way resulting from highway construction operations. It shall be the Contractor's responsibility that all workers and persons employed by the Contractor or the Contractor's agents, or under the Contractor's control, shall use due care that no person or property is injured, and that no rights are infringed in performing the work. The Contractor shall compensate injured parties.

The Contractor shall be responsible for traffic control persons or other personnel, equipment or procedures required for the protection of the railway trackage or property. Where it is found necessary to divert, relocate or re-route utilities or other facilities and infrastructure during the course of operations, or where it is found necessary to make repairs to such installations as a result of contract operations, such re-routing or repair work shall be the responsibility of the Contractor, and shall be for the Contractor's account.

Any costs borne directly by or billed directly to the Ministry which relate to the above responsibilities will be charged to the Contractor without any surcharge.

The Contractor shall restore and clean all slopes between the highway and the railway track together with any fences, ditches and drains disturbed by the work to the satisfaction of the Ministry Representative.

The Contractor shall give a minimum of thirty (30) days advance notice to the railway company, or other timelines specified by the railway, for any work being done adjacent to, on or above the rail right-of-way. Notice will include submission of detailed plans and procedures, as specified in the Contract, for review and approval, as required, by the railway company. No Work will be done (construction or demolition) without approval from the railway and consent from the Ministry Representative.

145.28.05 Construction Stakes, Monuments, Benchmarks and Tide Gauge – The Contractor shall set, as the Ministry Representative may direct and as is necessary for the proper layout of the Work, monuments and bench marks of concrete or other material sufficiently permanent and rigid to remain intact during the duration of the Work. The Contractor shall particularly avoid

destroying or disturbing such monuments or construction stakes and, should such destruction be unavoidable, shall notify the Ministry Representative in advance so that the points may be properly referenced and re-established.

145.28.06 Protection and Restoration of Property and Environment – The Contractor shall not enter upon private property for any purpose without first obtaining permission of the owner of that property and shall be responsible for the preservation of public property along and adjacent to the Work, and shall use every reasonable precaution necessary to prevent damage or injury thereto.

The Contractor shall be liable for all damages caused by fire started by the Contractor and shall, under no circumstances start fires without first securing the required permits and approval of authorities having jurisdiction even though it may be ordered or required to do such burning.

145.29 Haul Routes and Overloads – The Contractor shall be responsible for ascertaining that the Contractor's haul routes are acceptable to the authority having jurisdiction for highway loading.

Overweight or oversize vehicles are not permitted to use existing highway as a haul road for the hauling of construction materials, notwithstanding Section 7.04 of the Commercial Transport Regulation, as amended from time to time. Where a portion of the existing highway is to be later reconstructed as part of the Work, the Ministry Representative may authorize overweight or oversize haul on that portion of the highway, under such conditions as the Contract or the Ministry Representative may establish.

Off-highway equipment may be permitted to cross the existing highway only at specific locations as approved by the Ministry Representative.

Once the crushed granular base, surfacing or pavement work has commenced on any portion of the new, reconstructed or relocated highway, no overloads will be permitted on that portion unless written approval is given by the Ministry Representative.

145.30 Health and Safety – See also SS 135 Construction Site Safety.

145.30.01 Occupational Health and Safety Program – The Contractor shall provide an Occupational Health and Safety Program in accordance with SS 135.

145.30.02 Worksite Hazards – The Contractor shall erect and maintain signs, fences and barricades at the ends of the bridges before they are opened to traffic and at other dangerous areas of the Project, as necessary to prevent access of vehicles and pedestrians to the unfinished bridge and other dangerous areas.

145.30.03 Equipment – All trucks and other equipment used by the Contractor whether owned or rented for use on the Project shall meet manufacture's specifications for its purpose.

All trucks used on public roads shall have Motor Carrier approval.

Hire of trucks and other equipment paid on a production rate basis constitutes a Subcontract.

145.30.04 Audible Back Up Warning Device – All haul vehicles shall be equipped with an automatic audible warning device which will activate whenever the vehicle is backing up. The warning device shall be clearly audible above the ambient noise level at a minimum distance of 6 m from the back of the vehicle

145.31 Unacceptable Work – The Contractor shall correct promptly any Unacceptable Work to the satisfaction of the Ministry Representative.

145.32 Temporary Excavation – Where temporary excavations are undertaken in the course of the Work, the Contractor shall be responsible for maintaining the stability of adjacent ground and all structures and other works thereon, and for the cost of any temporary works which it provides for this purpose for which payment is not expressly provided in the Contract.

145.33 General Sign Requirements

145.33.01 Contractor Identification Signs – The Contractor may supply and install contractor identification signs from the time of occupying the Site until the Actual Completion Date. The Contractor shall maintain and remove such signs and all costs of supply, installation, maintenance and removal shall be the responsibility of the Contractor.

The contractor identification sign shall display only the Contractor's company name and beneath it the word "Information" followed by a telephone number which directly connects any caller to the Contractor's personnel who are familiar with, and are currently active on the Project.

The sign will not be reflectorized, and the background colour will not be either red, yellow or orange. The sign dimensions will not exceed 2,400 mm horizontal and 1,200 mm vertical, and the text size (height) will not be less than 150 mm.

The contractor identification sign will be erected only in conjunction with the C-35 project identification sign, and will not be in advance of or at the same location as the C-35 sign. Where there is no C-35 sign, no contractor identification sign will be permitted.

No contractor identification sign may be installed without prior approval of both the sign and the intended installation and location by the Ministry Representative who, in determining whether or not to grant such approval, will not necessarily be limited to the requirements set out above.

145.33.02 Prime Contractor Sign – When designated as the Prime Contractor, the Contractor shall provide "Prime Contractor" signs as specified in SS 135 Construction Site Safety.

145.33.03 Traffic Control Signs – See SS 194 Traffic Management for Work Zones.

145.34 Timelines for Review of Contractor Submissions – Timelines specified in the Contract for the Ministry to review Contractor submissions (for example: Quality Management Plans, Construction Environmental Management Plans, Bridge Erection and Demolition Plans) are, unless explicitly specified otherwise, target times.

The timelines specified are considered by the Ministry to be reasonable under most circumstances, and typically allow for one full review by the Ministry and third-party approval agencies (such as environmental Ministries and utilities), plus a short confirmatory review of a revised submission to ensure all issues raised have been adequately addressed.

However, actual time required for the Ministry and agencies to complete the review can vary significantly based on any number of factors, including without limitation:

(a) the quality of the original submission;

(b) the number of times a document must be resubmitted before it is judged compliant;

(c) the time taken by the Contractor while revising a submission;

(d) the number of working days within the review period; and

(e) the effect of any reduced staff availability during holiday and vacation periods.

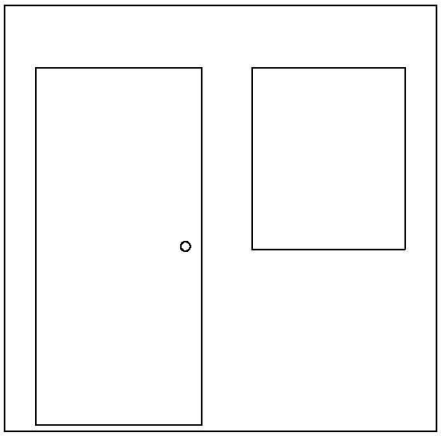
The Contractor is encouraged to consult with the Ministry Representative, based on the anticipated schedule of the Contractor's submissions, and plan its actual submission times accordingly.

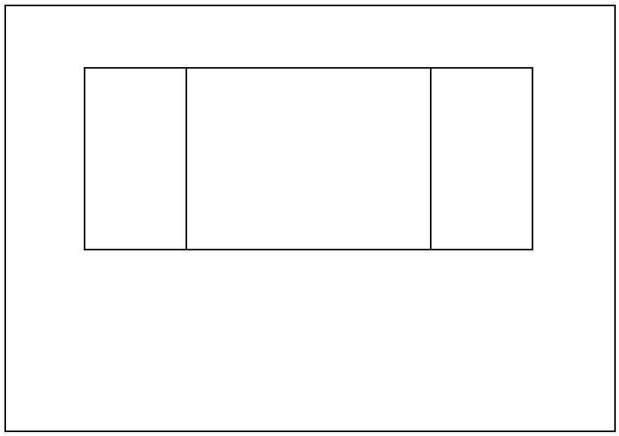
APPENDIX 145-A

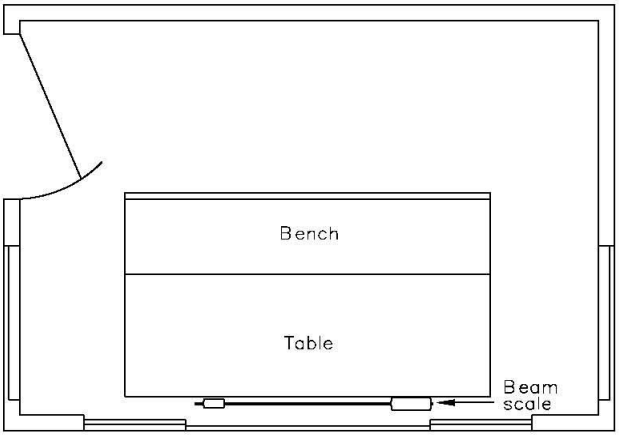
MINISTRY OFFICE REQUIREMENTS

ELEMENT	OFFICE TYPE		
	A	**B**	**C**
Lockable desk/chair	1	2	3
Typist desk/chair	-	-	1
Drafting table 1.2 m x 2.4 m /stool	1	1	1
Reference table 1.2 m^2	-	1	3
Conference table 1.2 m x 2.4 m	-	-	1
Additional chairs	2	2	9
Plan case or drawing storage unit	1	1	2
Filing cabinet (# drawers)	1 – 2 drawers	1 – 4 drawers	3 – 4 drawers
Shelving bookcase m^2	2	5	20
Fire Extinguisher for Class A, B and C fires	1 – 5 kg	1 – 5 kg	2 – 5 kg
Supply Activated Telephones and Lines (if cellular cover is unavailable or poor)	One (1)	One (1)	Three (3) on two (2) lines with external bell and switch
High Speed Internet	Wireless covering entire office or one drop at each desk and drafting table		
Toilet	On site	On site	In washroom, hot and cold running potable water and wash basin
First Aid Kit	On site	#1 Unit Kit	#1 Unit Kit
HVAC	As needed to maintain an indoor temperature of 20°C to 27°C year-round.		
Floor Areas – m^2			
a) Entrance	-	-	5
b) Ministry Representative's office	12	12	12
c) Additional Staff office	-	8	20
d) Wash room	-	-	As required
e) Store room	-	-	5

SCALE HOUSE	SP145-01







NOTES:

1. MINIMUM SIZE IS 2100 mm BY 3000 mm
2. INSULATED AND VENTILATED
3. ADEQUATELY HEATED WHEN SHADE TEMPERATURE IS BELOW 16°C

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

APPENDIX 145-B

Survey Layout Services and Products[1]

Survey Layout	Maximum Interval	Product	Tolerances[2]
Right-of-way[3]	At each point of deflection and at sufficient points between as to be continuously visible.	Stake showing station and offset, or flagging.	Sufficient accuracy to prohibit encroachment into adjoining properties.
Clearing and Grubbing	Same as Right-of-Way.	Same as Right-of-Way.	Sufficient accuracy to prohibit encroachment into adjoining properties.
Grading - Slope Stakes	10 m in rock cuts; 20 m in all other cases. For Machine Controlled Grading Projects – only an orientation stake is required every 100 m	One slope stake each side, at top of cut or bottom of fill, showing station, offset, vertical dimension to subgrade, and slope, plus cut/fill transition stake. Non-standard ditches will be staked separately. An additional slope stake, where applicable, at the top of a rock cut after the removal of overburden.	Up or down chainage: ±0.30 m Offset from ℄: ±30 mm Vertical: ±25 mm
Grading – Subgrade	20 m. For Machine Controlled Grading Projects – only an orientation stake is required every 100 m	One stake at each side of the subgrade, showing station, offset and grade at the stake location, one at each break point, and one at centreline.	Up or down chainage: ±0.30 m Offset from ℄: ±30 mm Vertical accuracy per SS 201.47
Top of Sub-base	20 m. For Machine Controlled Grading Projects – only an orientation stake is required every 100 m	One stake at each side of the sub-base course, showing station, offset and grade at the stake location, one at each break point, and one at centreline.	Up or down chainage: ±0.30 m Offset from ℄: ±30 mm Vertical accuracy SS 202.25.04
Each Base Course	20 m. For Machine Controlled Grading Projects – only an orientation stake is required every 100 m	One stake at each side of the base course, showing station, offset and grade at the stake location, one at each break point, and one at centreline.	Up or down chainage: ±0.30 m Offset from ℄: ±30 mm Vertical accuracy per SS 202.26.04

[1] This table shows layout details for general situations; particular circumstances may require more or less staking.

[2] Staking tolerances also represent the tolerances for construction of the Work. Any Work constructed outside of these tolerances constitute a "change" and must be approved by the Ministry Representative and shown on the Record Drawings. See Special Provision clause titled "Marked Up Drawings".

[3] The right-of-way limit will be laid out only where there is work, including utility relocation, to be performed within three metres of it.

Survey Layout	Maximum Interval	Product	Tolerances[2]
Final Base Course only	10 m. For Machine Controlled Grading Projects – only an orientation stake is required every 100 m	"Blue tops"[4] at each break point across the base course surface, or final grade stakes.	Up or down chainage: ±0.30 m Offset from ℄: ±30 mm Vertical accuracy per SS 202.26.04
Culverts	Inlet and outlet.	One stake at each end of the culvert, plus an offset line, showing invert elevation.	Up or down chainage: ±0.30 m Offset from ℄: ±30 mm Vertical: ±20 mm
Storm Drainage, Subdrain, Watermain or Sanitary Sewer		Stakes showing locations of manholes, catch basins and other structures, and invert locations of pipe inlets and outlets.	Up or down chainage: ±0.30 m Offset from ℄: ±30 mm Vertical: ±20 mm
Retaining Walls	Not more than 10 m, and at alignment changes.	One stake showing control line location and either the elevation at the top of wall or the elevation at the bottom of footing excavation.	As defined in the Special Provisions.
Paving	100 m on tangent, 20 m on curves, and at each deflection point.	Reference points.	N/A
Concrete Barriers	Same as paving.	Same as paving.	Offset from ℄: ±30 mm
Signs	Stake at each sign location.		Up or down chainage: ±0.30 m Offset from ℄: ±30 mm
Curb and Gutter	10 m and at alignment changes. Curb returns: 5 m or at quarter points, whichever is less.	Offset hub and nail with cut/fill to gutter grade.	Up or down chainage: ±0.30 m Offset from ℄: ±30 mm Vertical: ±10 mm
Median/Island Curb	Continuous.	Paint line at face/edge of curb.	Up or down chainage: ±0.30 m Offset from ℄: ±30 mm
Sidewalk or Path	5 m and at alignment changes. No layout required if parallel to curb.	Offset stake or nail with cut/fill to grade.	Up or down chainage: ±0.30 m Offset from ℄: ±30 mm Vertical: ±15 mm

[4] "Blue top" - a stake driven so that its top surface is at the required elevation for the finished surface of the work.